COVID-19 PANDEMIC STOCK MARKET REACTIONS- VISUAL ENTERTAINMENT INDUSTRY

This study examines stock market reactions in the visual entertainment industry during the first and second waves of COVID-19. Using stock price and trading volume data from 15 firms, the research analyzes correlations, trends, and comparative differences through Power BI visualizations.

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Table of Contents

Introduction

Background of the Research

The visual entertainment industry, encompassing film production, television, streaming services, and cinema chains, plays a significant role in the global economy. Prior to 2020, the industry was experiencing robust growth, largely driven by the expansion of digital streaming services such as Netflix and Disney+, alongside a strong global film market. Consumers increasingly shifted toward digital content consumption, fueled by advancements in mobile and internet connectivity, leading to rising demand for on-demand services.

However, the COVID-19 pandemic in 2020 disrupted this growth trajectory, causing major shifts in consumer behavior and business operations. Lockdowns, social distancing measures, and the closure of theaters and live event venues led to a seismic shift in the entertainment landscape. While streaming services witnessed a surge in demand as people turned to digital content during lockdowns, traditional sectors like cinema chains and live events faced substantial losses due to restrictions. This divergence in impact across different segments of the entertainment industry makes it crucial to analyze stock market reactions and investor behavior during this period.

Study Rationale

The COVID-19 pandemic reshaped industries globally, with the entertainment sector—particularly visual entertainment—experiencing both challenges and opportunities. The combination of consumer-driven demand, technological adaptation, and capital investment in this sector makes its market performance a critical indicator of economic resilience during crises.

While some firms benefited from increased digital consumption, others, particularly cinema chains and live event organizers, suffered due to lockdown-induced losses. This research seeks to analyze how COVID-19 affected stock performance in the visual entertainment industry by focusing on adjusted close prices and trading volumes. The findings will help investors and stakeholders understand stock market behavior in times of economic disruption.

Research Objectives

This study aims to analyze the stock market performance of the visual entertainment industry during the first and second waves of COVID-19. Specifically, it aims to:

- Examine fluctuations in adjusted close prices and stock trading volumes across 15 selected firms during the pandemic.
- Identify significant differences between stock market performance in the first wave (early 2020) and the second wave (late 2020).
- Provide data-driven insights into investor behavior and market resilience during a global crisis.

Research Questions

- **RQ1:** What is the impact of COVID-19 on the trading volume of visual entertainment stocks?
- **RQ2:** What is the difference between stock market adjusted close prices and stock market volume during the first and second waves of COVID-19?

Significance of Research

This study contributes to the field of financial investment during global crises by offering insights into stock market behavior within the entertainment industry.

- **Theoretical Contribution:** The study enhances understanding of market dynamics during crises, providing empirical evidence on stock price behavior and trading volumes in a specific industry.
- **Practical Contribution:** Investors, portfolio managers, and financial decision-makers can use these findings to develop risk management strategies and make informed

investment decisions during future economic shocks. Additionally, corporate managers in the entertainment industry can use these insights to evaluate financial resilience strategies and sustain business operations amid crises.

The Innovativeness of the Project

Unlike industries such as travel and hospitality, which faced uniform declines, the entertainment sector showed a bifurcated response—streaming platforms thrived while cinema chains struggled. This project takes a unique approach by focusing specifically on the visual entertainment industry, analyzing stock price movements and trading volumes to understand investor behavior.

By comparing streaming services and cinema chains, this study examines resilience, adaptation strategies, and financial performance throughout the first and second waves of COVID-19. By leveraging historical stock data and trend analysis, we aim to provide valuable insights that can guide future investment strategies and industry preparedness for economic disruptions.

Literature Review

Overview of the COVID-19 Pandemic and Its Impact on the Stock Market

The COVID-19 pandemic had an unprecedented effect on global stock markets. Baker et al. (2020) highlight that the pandemic's impact on financial markets was far more severe compared to previous infectious disease outbreaks due to the modern economy's heavy reliance on service sectors, including entertainment. The pandemic-induced lockdowns and travel restrictions led to heightened market volatility, significantly affecting industries dependent on physical presence, such as film production and cinema chains.

Zhang et al. (2022) empirically demonstrated that COVID-19 substantially reduced stock market returns within cultural and entertainment industries. However, they found that digitization played a crucial role in mitigating losses. Companies operating in digital entertainment, such as streaming platforms, performed better than those relying on traditional revenue models.

Historical Global Crises and Their Impact on Stock Markets

To better understand COVID-19's effect, it is essential to compare it with past crises that disrupted financial markets:

1. **The 2008 Global Financial Crisis**: The global recession led to a dramatic decline in stock market indices worldwide, causing long-term instability in industries such as banking, real estate, and media. Entertainment companies faced reduced advertising revenues and lower consumer spending on non-essential entertainment.
2. **The Dot-Com Bubble (2000-2002)**: The collapse of overvalued tech stocks had a significant spillover effect on the broader economy, impacting entertainment firms reliant on digital media investments.
3. **The Great Depression (1929)**: One of the most severe financial crises in history, it led to significant disruptions in Hollywood's early film industry, with reduced consumer spending on cinema and entertainment.

These crises illustrate how economic downturns impact market behavior and industry resilience. In contrast, COVID-19's impact was unique in that it created a divergence—while some sectors suffered, digital entertainment thrived.

COVID-19's Impact on the Entertainment Industry

COVID-19's effect on the visual entertainment industry was profound, leading to shifts in consumer habits, disruptions in production, and stock market volatility.

- **Streaming Services and Digital Entertainment**: Luo (2020) referred to this period as the "streaming war," where platforms like Netflix, Disney+, and Amazon Prime saw a surge in subscribers. This unprecedented growth highlighted how digital adaptability helped companies withstand economic downturns.
- **Traditional Cinemas and Live Events**: Shah et al. (2020) emphasized the challenges faced by cinemas, including halted film production, postponed movie releases, and drastically reduced box office revenues. However, their research also noted that the Chinese film industry rebounded faster than other markets, illustrating regional differences in recovery.

The Next Normal for the Visual Entertainment Industry

The post-pandemic entertainment industry is expected to see lasting transformations:

- **Hybrid Entertainment Consumption**: The combination of digital and in-person experiences is likely to shape future entertainment trends. Streaming services will continue to dominate, while cinemas may need to adopt innovative approaches such as hybrid releases and premium experiences to attract audiences.

- **Investor Strategies in a Post-COVID World**: Ryu and Cho (2022) proposed a framework for understanding pandemic-induced changes, categorizing businesses into those that thrived due to digital integration and those that struggled due to reliance on physical locations. This framework can help investors evaluate long-term market trends.

Methodology

Overview of the Methodology

This study examines stock market reactions in the visual entertainment industry during the first and second waves of the COVID-19 pandemic (January–December 2020). The methodology encompasses data collection, cleaning, analysis, and storage using Python for data extraction and preprocessing, followed by visualization and further analysis in Power BI.

Data Collection Process and Technique

1. Source of Data Collection

The stock market data was sourced from Yahoo Finance using the yfinance Python library.

Rationale for Yahoo Finance:

- Provides free and reliable historical stock market data.
- Easily integrates with Python for automated data extraction.
- Includes key stock metrics such as Open, High, Low, Close, Adjusted Close, and Volume.

2. Analytical Tools Used for Data Extraction

- **Python & yfinance**: Extracted daily stock prices for multiple companies.
- **Pandas**: Processed and structured the extracted data.

3. Time Frame & Justification

The study divides the one-year period into two waves:

- **First Wave:** January 1, 2020 – June 30, 2020
- **Second Wave:** July 1, 2020 – December 31, 2020

This segmentation allows for a comparative analysis of stock performance before and after major pandemic-related economic disruptions.

4. Types of Data Collected

- **Numerical Data:**
  - Stock Prices (Open, High, Low, Close, Adjusted Close)
  - Trading Volume (total number of shares traded daily)
- **Categorical Data:**
  - Wave Classification (First vs. Second Wave)
  - Stock Ticker Symbols (Company identifiers)

5. Stock Market Indexes Used

While the dataset primarily focuses on stock price movements, the study may incorporate broader market indices such as:

- **S&P 500**: Represents the top 500 US companies.
- **NASDAQ Composite**: Includes tech and entertainment stocks.
- **Dow Jones Industrial Average (DJIA)**: Covers large US corporations.

Data Cleaning Process

After extracting the stock market data, it was saved as a CSV file (Stock_Market_Data.csv) for further analysis. The dataset was then imported into Power BI for data processing, where several cleaning and transformation steps were applied:

1. **Handling Missing Values:** Missing stock prices and trading volumes were either filled using forward-fill/backward-fill methods or dropped when necessary.
2. **Feature Engineering:** Additional derived columns and calculated measures were created to enhance analysis.
3. **Date Formatting:** Converted string-based dates into datetime format for accurate time-series analysis.
4. **Ensuring Consistency:** Verified that all 15 companies had complete data across both the first and second waves.
5. **Categorization of Companies:**
   a. A new column (CompanyCategory) was created using a DAX formula to classify firms into "Streaming" or "Traditional" entertainment.
6. **Stock Market Data for 15 Firms in the Visual Entertainment Industry**
   The study focuses on the following 15 companies:
   - **Streaming Services:** Netflix (NFLX), Disney (DIS), Warner Bros. Discovery (WBD), Roku (ROKU), Sony (SONY), Warner Music Group (WMG)
   - **Traditional Entertainment:** Comcast (CMCSA), Cinemark Holdings (CNK), IMAX (IMAX), Fox Corporation (FOXA), Live Nation Entertainment (LYV), TKO Group (TKO), Lionsgate (LGF-B), Liberty Media (FWONA), Paramount Global (PARA)
7. **Correlation Analysis:**

a. A correlation measure was defined in Power BI using DAX to evaluate the relationship between adjusted closing prices and trading volume.

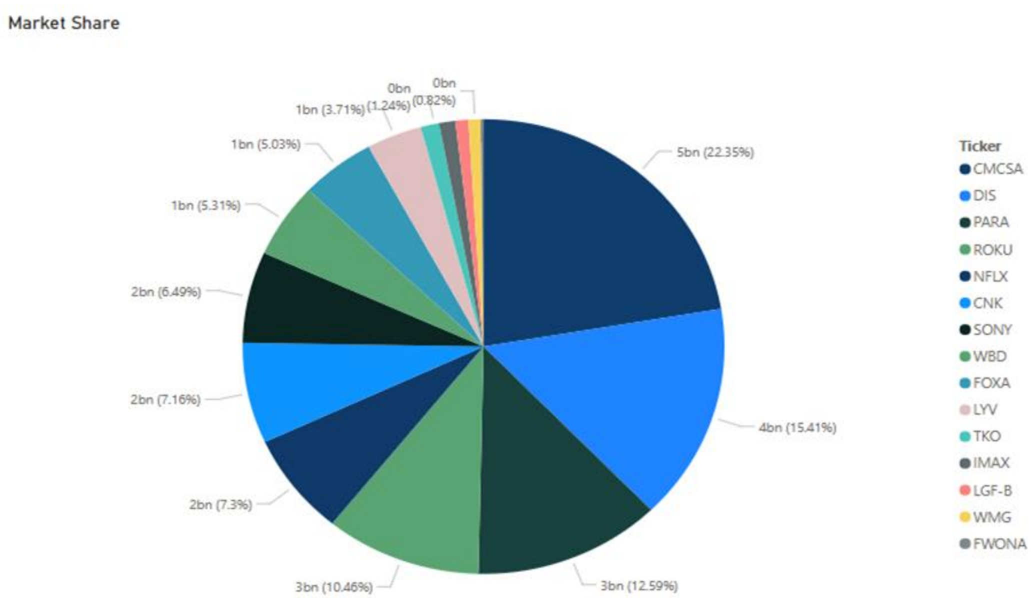
8. **Stock Market Aggregates:**
    a. Summarized industry-wide stock market performance using aggregate measures such as Industry_Total_Close.

Data Storage

The cleaned and transformed dataset was stored as a CSV file, which was then used for visualization and analysis in Power BI. The structured data format allowed for efficient trend analysis and insight generation related to stock market behavior during the pandemic.

Overview of Data Analysis

This study analyzes the stock market reactions of the visual entertainment industry during the first and second waves of the COVID-19 pandemic. Data from Yahoo Finance was used to examine adjusted close prices and trading volumes for 15 firms, categorized into streaming and traditional entertainment companies. The analysis was conducted using Power BI and included correlation analysis, trend analysis, and bar chart comparative analysis to identify significant market shifts. The findings provide insights into how stock prices and trading volumes evolved during these periods.



Correlation Analysis

Importance of Correlation Analysis

Correlation analysis measures the strength and direction of relationships between different stock market variables. In this study, it helps determine the degree to which adjusted close prices and trading volumes of entertainment industry stocks move together. Understanding these correlations allows investors to identify potential market dependencies, risk diversification opportunities, and sector-wide trends during economic disruptions like the COVID-19 pandemic.

Methodology

The correlation analysis was conducted using Power BI, where correlation coefficients were calculated for different scenarios:

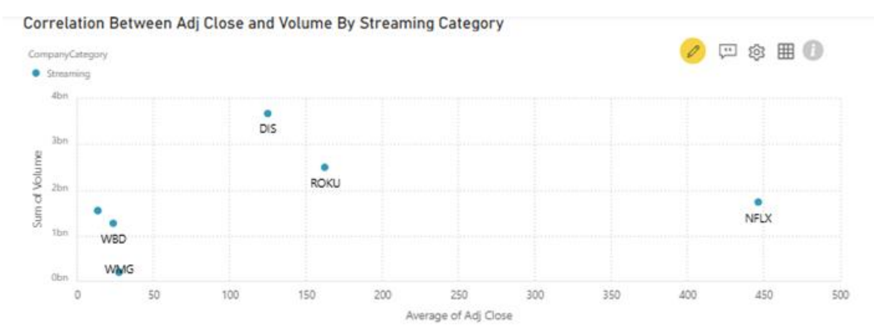
- **Overall correlation**: Examining relationships across the entire dataset.
- **Correlation by wave**: Analyzing how stock relationships varied between the first and second waves of the pandemic.
- **Correlation by category**: Comparing correlation patterns between streaming services and traditional entertainment firms.
- **First wave vs. second wave within each category**: Investigating whether correlations within streaming and traditional firms changed between waves.

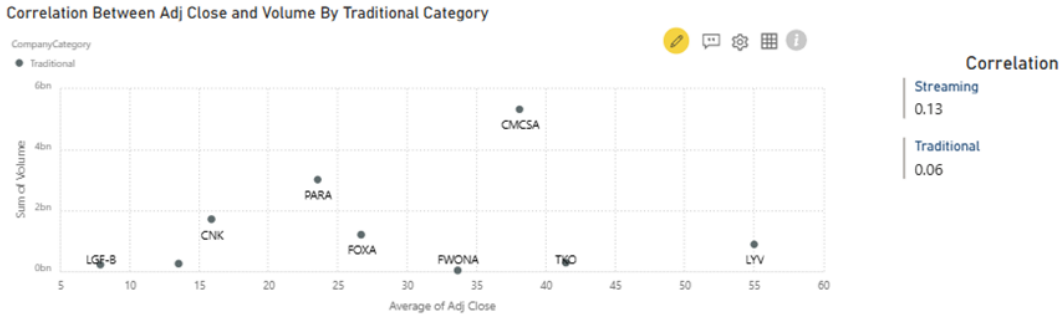
Key Findings

- **Overall Correlation**: Adjusted close prices showed strong positive correlations among firms in the same industry segment, indicating that stock movements within the visual entertainment sector followed similar trends.

- **Correlation by Wave**:

  - During the first wave, correlations were weaker, suggesting increased market uncertainty and independent stock movements.
  - The second wave saw stronger correlations, indicating a more synchronized market reaction as investors adjusted to the pandemic's economic effects.
- **Correlation by Category**:





- Streaming stocks had a higher degree of correlation, possibly due to uniform demand increases for digital entertainment.
- Traditional entertainment stocks exhibited mixed correlations, reflecting varying recovery rates among companies.
- Streaming stocks showed more consistent correlations across both waves, implying steady investor confidence.
- Traditional entertainment stocks displayed a shift from weaker correlations in the first wave to stronger correlations in the second wave, suggesting a clearer market consensus on the sector's financial outlook.

Trend Analysis

Importance of Trend Analysis

Trend analysis identifies patterns in stock price movements over time, allowing for an understanding of market performance and investor sentiment shifts. This study focuses on trends in adjusted close prices and trading volumes across the first and second waves of COVID-19 to reveal how the industry adapted to economic disruptions.

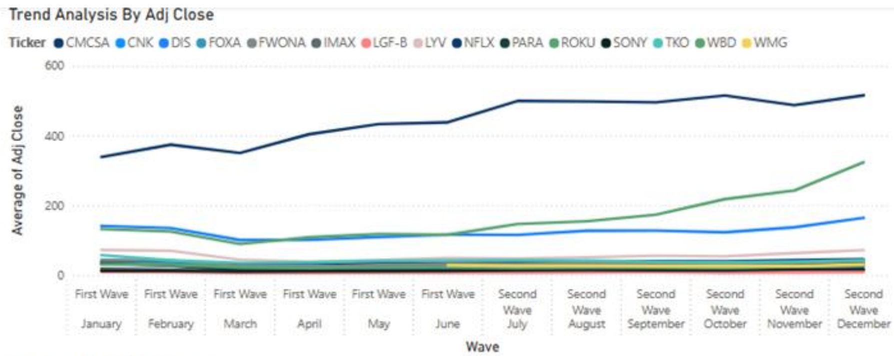
Methodology

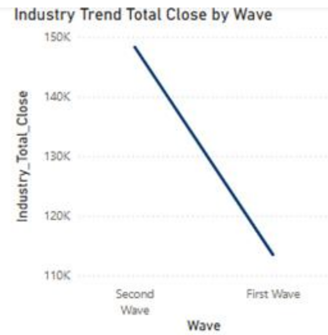
Power BI was used to visualize stock trends through

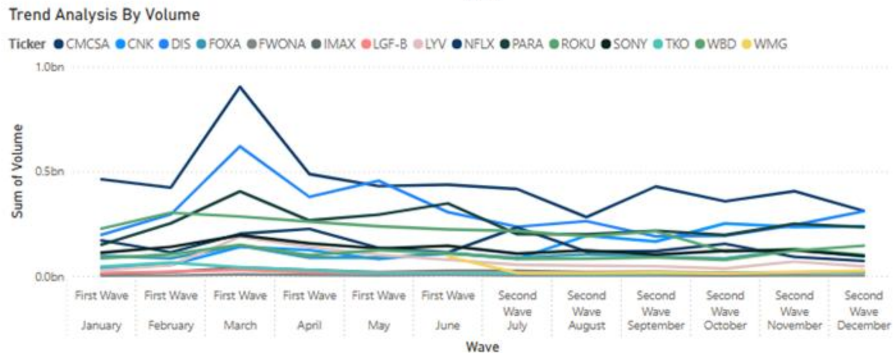
- **Overall period trend analysis**: Tracking adjusted close prices and volumes over the entire year.
- **Monthly trend analysis by wave**: Summarizing stock market movements in each pandemic wave.
- **Monthly trend analysis by category**: Comparing streaming and traditional entertainment trends.

Key Findings

- **Overall Trends**:

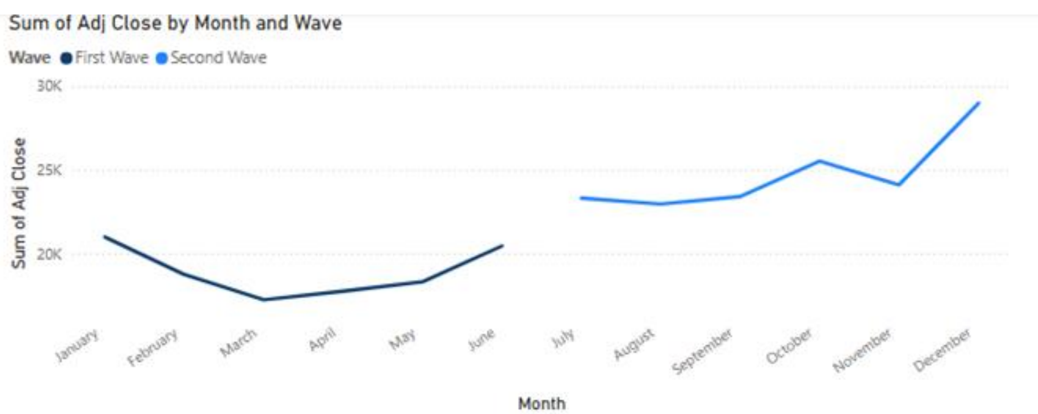


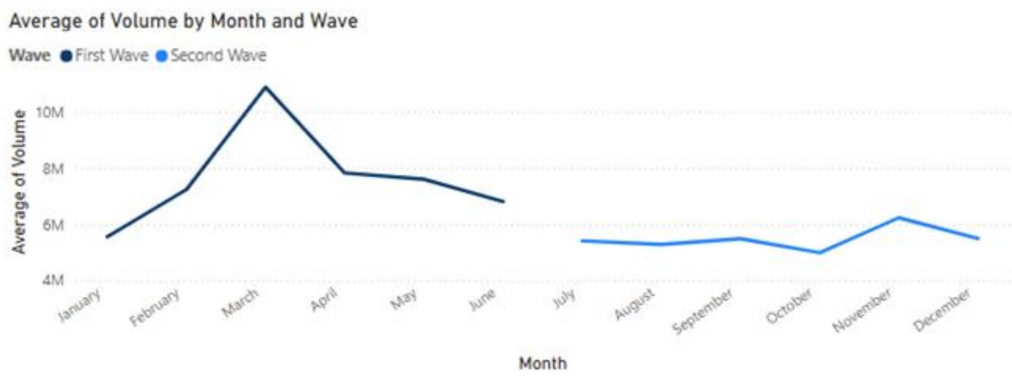




  - Adjusted close prices showed a strong upward trend for streaming firms, reflecting investor confidence in digital content consumption.
  - Traditional entertainment stocks initially declined but saw partial recovery in the second wave, aligning with reopening efforts.

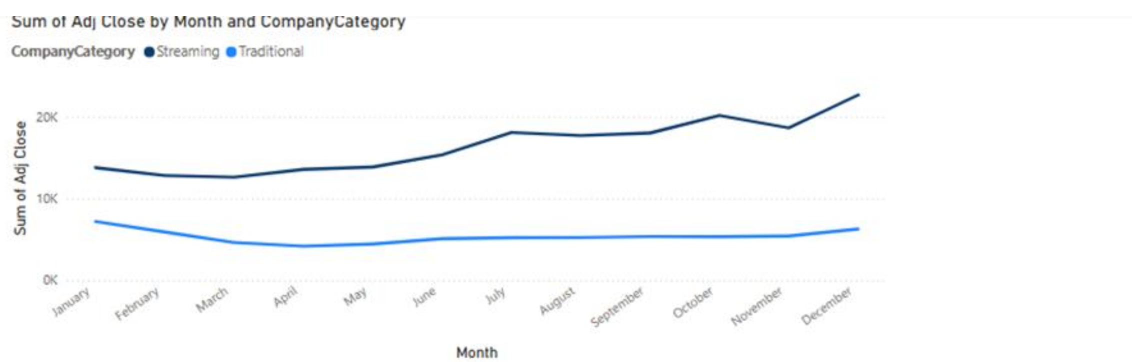
- **Adjusted Close Price and Volume Trends by Month & Wave**:





- The first wave exhibited high volatility, with price drops in early 2020 followed by recoveries as investors identified market opportunities.
- The second wave showed a more stable trend, with stock prices rising steadily as companies adapted to pandemic-related changes.

- **Adjusted Close Price and Volume Trends by Category**:



- Streaming firms experienced sustained price increases and stable trading volumes.
- Traditional entertainment firms saw a decline in trading volume, indicating reduced investor activity and lower confidence in recovery.

Comparative Analysis (Wilcoxon Signed Ranks Test Alternative)

Importance of Comparative Analysis

A comparative analysis assesses differences in stock market performance between the first and second waves. Since IBM SPSS was not used, Power BI visualizations serve as an alternative approach to understanding these variations.

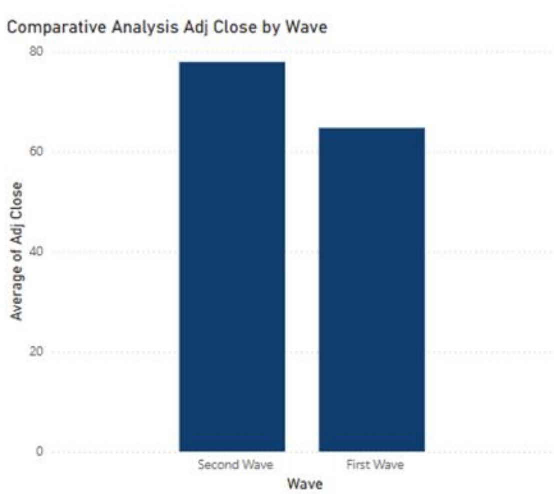
Methodology

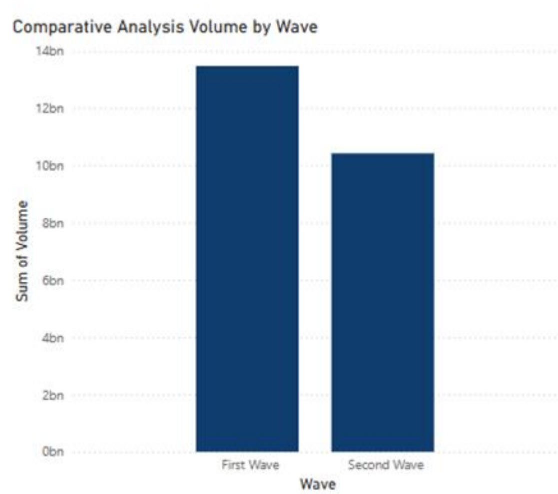
Visual comparisons were conducted using Power BI to evaluate:

- **Adjusted close prices and trading volumes by wave**.
- **Adjusted close prices and trading volumes by wave and category**.

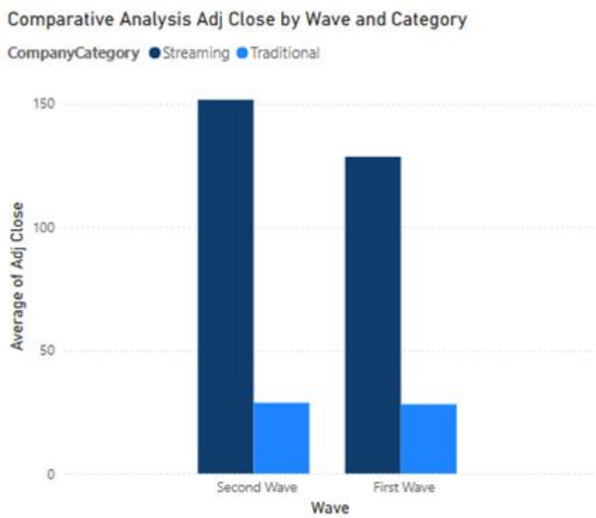
Key Findings

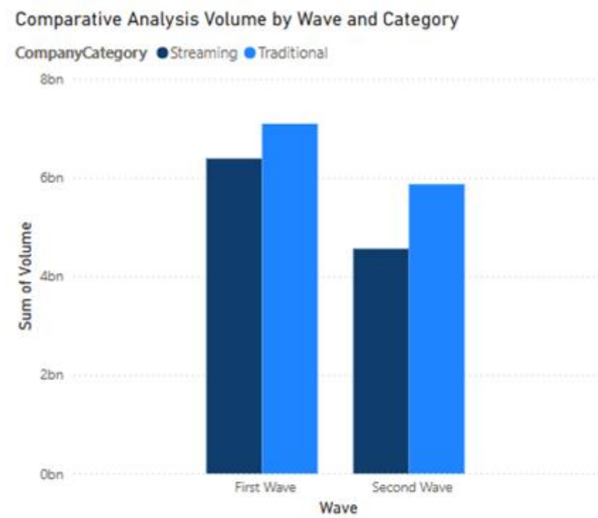
- **Adjusted Close Price and Volume by Wave**:





  - A notable increase in adjusted close prices in the second wave suggests improved investor confidence and industry stabilization.
  - Trading volumes varied significantly, with higher activity in the first wave as investors reacted to market uncertainties and lower activity in the second wave as trading stabilized.
- **Adjusted Close Price and Volume by Wave & Category**:





- Streaming firms showed consistently higher adjusted close prices in both waves, reinforcing their resilience during the pandemic.
- Traditional entertainment firms had lower adjusted close prices in the first wave but demonstrated a slow recovery in the second wave, possibly due to gradual reopening strategies.

Research Implications and Final Considerations

Research Implications

Significance of Findings

This study provides crucial insights into stock market reactions within the visual entertainment industry during the first and second waves of COVID-19. The correlation analysis reveals that market uncertainty in the first wave led to weaker relationships among stock prices, whereas the second wave demonstrated stronger correlations as investors adapted. Trend analysis highlights the resilience of streaming firms compared to traditional entertainment companies, emphasizing digital transformation as a key factor. The comparative analysis further supports these findings, showing improved investor confidence and stabilization in the second wave.

New Insights

- Streaming firms exhibited higher and more stable correlations, reinforcing their strong market position during economic disruptions.
- Traditional entertainment firms initially suffered but showed signs of recovery in the second wave, suggesting long-term market adaptation.
- Investor behavior became more predictable over time, with reduced market volatility and improved confidence during the second wave.

Practical Recommendations

1. **Investment Strategies:** Investors should consider streaming stocks as safer assets during economic downturns due to their resilience.
2. **Business Diversification:** Traditional entertainment firms should expand their digital offerings to mitigate risks in future crises.
3. **Crisis Preparedness:** Companies must develop adaptive financial strategies to handle periods of economic uncertainty effectively.
4. **Market Monitoring:** Decision-makers should utilize real-time data analytics to identify stock market trends and adjust strategies accordingly.

Theoretical Recommendations

1. **Sector-Specific Correlation Studies:** Future research should explore how external shocks impact different industry segments.
2. **Pandemic Response in Financial Markets:** Scholars should examine investor behavior changes across multiple economic crises.
3. **Longitudinal Analysis:** A study spanning multiple years would help determine whether post-pandemic recovery trends are sustained.
4. **Machine Learning in Stock Prediction:** Advanced modeling techniques could improve stock movement predictions during volatile periods.

Study Limitations

- **Data Constraints:** The study is limited to 15 firms, which may not represent the entire industry.
- **Market Volatility:** Unforeseen external factors, such as government policies, may have influenced stock trends beyond COVID-19.
- **Power BI Limitations:** While useful for visualization, Power BI lacks statistical testing capabilities like SPSS, affecting the depth of comparative analysis.
- **Timeframe:** The study focuses on two pandemic waves; a longer time horizon would provide deeper insights into sustained industry trends.

Conclusion

This study highlights the differentiated stock market behaviors of streaming and traditional entertainment firms during COVID-19. Streaming firms demonstrated resilience and strong investor confidence, whereas traditional entertainment firms faced initial setbacks but later showed signs of recovery. Correlation trends shifted from weak in the first wave to stronger in the second wave, reflecting increased market predictability. These insights provide valuable guidance for investors and industry stakeholders on responding to economic crises and adapting to market disruptions.

Abstract

This study examines stock market reactions in the visual entertainment industry during the first and second waves of COVID-19. Using stock price and trading volume data from 15 firms, the research analyzes correlations, trends, and comparative differences through Power BI visualizations. Key findings reveal that streaming firms exhibited strong resilience, maintaining positive price trends and stable investor confidence, whereas traditional entertainment firms struggled initially but showed recovery signs in the second wave. Correlation analysis indicates weaker relationships in the first wave due to market uncertainty, while the second wave demonstrated stronger correlations as investor behavior stabilized. Trend analysis confirms sustained growth in streaming stocks, whereas traditional entertainment stocks experienced fluctuations. Practical recommendations emphasize digital transformation and real-time market monitoring, while theoretical contributions suggest further research into sector-specific correlations and advanced predictive modeling. These findings contribute to a better understanding of market dynamics during economic crises, offering actionable insights for investors and industry stakeholders.

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